CURRICULUM VITAE

CLIFFORD ONYANGO

AI SOFTWARE ENGINEER | MACHINE LEARNING DEVELOPER | FULL STACK ENGINEER

PROFESSIONAL PROFILE

Computer Science student and emerging AI Software Engineer with practical experience developing software solutions across artificial intelligence, machine learning, full-stack development, data science, and technology-for-impact projects.

Demonstrates strong independent learning ability with a consistent pattern of transforming ideas into functional technical systems through research, experimentation, implementation, debugging, and continuous improvement.

CORE TECHNICAL AREAS

- Artificial Intelligence
- Machine Learning
- Computer Vision
- Data Science
- Full-Stack Web Development
- Backend API Development
- Database Systems
- Technology-for-Impact Projects

TECHNICAL SKILLS

Skill / Technology	Level	Recruiter Assessment
Python	Intermediate	Primary language for AI, data science, automation, and backend development.
JavaScript	Intermediate	Frontend development and interactive web applications.
HTML/CSS	Intermediate	Responsive web development and UI implementation.
SQL	Developing-Intermediate	Database management and application data systems.
C/C++	Developing	Programming fundamentals, systems, and embedded exploration.
React	Intermediate	Component-based frontend development.
Machine Learning	Intermediate	Data preparation, feature engineering, model development, and evaluation.
Computer Vision	Developing-Intermediate	Image processing and satellite imagery analysis.
Git/GitHub	Intermediate	Version control and collaborative development.
Docker	Developing	Containerization and development environment management.
REST APIs	Intermediate	Backend application integration.
Cloud Deployment	Developing	Early exposure requiring further professional development.

AI & DATA SCIENCE TECHNOLOGIES

- Pandas
- NumPy
- Scikit-learn
- TensorFlow
- Matplotlib
- Streamlit
- Feature Engineering
- Predictive Modelling
- Dataset Preparation
- Model Evaluation

EDUCATION

Bachelor of Computer Science

Zetech University

SELECTED PROJECT EXPERIENCE

DigiTeach — Digital Education Platform

A digital education platform focused on improving access to technology learning resources.

Key Contributions:

- Developed responsive user interfaces.
- Translated design concepts into functional web experiences.
- Structured frontend application components.
- Managed project assets and user interface elements.
- Applied user-focused product development principles.

Technologies: React, JavaScript, HTML, CSS, frontend development tools.

AI Environmental Monitoring System

An applied artificial intelligence project exploring machine learning and computer vision techniques for environmental analysis using image-based and structured datasets.

Key Contributions:

- Prepared and processed datasets.
- Performed feature extraction.
- Developed machine learning workflows.
- Experimented with predictive models.
- Analyzed environmental data patterns.

Technologies: Python, machine learning libraries, computer vision tools, data processing frameworks.

Machine Learning Prediction Systems

Developed experimental predictive analytics solutions involving dataset preparation, model training, evaluation workflows, and prediction pipelines.

- Processed datasets.
- Tested machine learning algorithms.
- Evaluated model performance.
- Built experimental AI workflows.

INDEPENDENT SOFTWARE DEVELOPMENT

Designed, developed, and maintained multiple independent technical projects involving artificial intelligence, software applications, data analysis, web development, automation, and experimentation with emerging technologies.

Engineering activities include:

- Requirement analysis
- Software design
- Implementation
- Debugging
- Testing
- Deployment experimentation
- Technical research

PROFESSIONAL STRENGTHS

Technical Learning Ability: Ability to independently acquire unfamiliar technologies and apply them to practical projects.

Problem Solving: Strong ability to break complex challenges into manageable technical solutions.

Product Thinking: Focuses on solving meaningful problems rather than only implementing technology.

Adaptability: Comfortable working across multiple technical domains.

Project Ownership: Demonstrates initiative in taking projects from ideas to working implementations.

CURRENT DEVELOPMENT AREAS

- Advanced software architecture
- Automated testing
- Cloud engineering
- MLOps
- Open-source collaboration
- Large-scale system design
- Production monitoring

CAREER OBJECTIVE

To contribute as an AI Software Engineer or Software Developer by building reliable intelligent systems that combine machine learning, software engineering, and user-focused product development.

Seeking opportunities where technical curiosity, problem-solving ability, and continuous learning can contribute to impactful technology solutions.

RECOMMENDED PROFESSIONAL POSITIONING

AI-focused Software Engineer capable of building intelligent systems from data preparation and model development through API integration, user interfaces, and deployment.